## VOEDING 12 V

J1 — 12V IN (GND, VIN_RAW)

Q1 — AO4407A (GS_G)

D2 — BZT52C15 (V12 → GS_G)

D1 — SMBJ15CA (V12 → GND)

C24 330µ 25V · C25 330µ 25V · C26 100µ 25V · R36 100k · R37 100k

C27 100µ 50V

## BUCK 12 V –> 5 V 5 A

U1 — TPS54560B (VIN, EN, COMP, RT/CLK, GND, BOOT, SW, FB)

C28 10µ 25V · C29 10µ 25V · C30 10n · C31 47µ 25V · R38 100k · R39 24k · R40 10R · C32 68u

D4 — B560C (V12 → GND)

C33 47µ 50V · C34 47µ 50V · C35 47µ 50V · R41 51k · R42 10k · R43 100k · C36 · C37 · C38 · C39

D5 — groen 5V (GND ← LED5_A)

R44 1k

J13 — 5V PI (GND, V5)

## 3,3 V (ESP32 + IMU)

D6 — SS14 (V5_ESP → V5)

D7 — SS14 (V5_ESP → VBUS_ESP)

U2 — AMS1117-3.3 (V5_ESP → V3V3)

C40 22µ · C41 100n · C42 10µ

## ESP32-S3 (SERVO-BUS + IMU + USB NAAR PI)

U3 — ESP32-S3-WROOM-1-N16R8

R45 10k (ESP_EN) · R46 10k (ESP_IO0) · R47 10k (LED_ESP_R)

D8 — blauw ESP (GND ← LED_ESP_A)

C43 100n · C44 100n · C45 10µ

R48 1k (LED_ESP_A) · R49 1k (LED_A) · R50 1k (LED_B)

SW1 — RESET (ESP_EN, GND)

SW2 — BOOT (ESP_IO0, GND)

## IMU MPU-6050

U4 — MPU-6050 (VLOGIC, SDA, SCL, ADO, FSYNC, AUX_DA, AUX_CL, INT, CPOUT, REGOUT, VDD, GND)

C46 100n · C47 2µ · C48 100n · C49 2µ · C50 100n · C51 2µ

## USB-C NAAR PI 4

J8 — USB-C (VBUS, CC1, CC2, D-, D+, SBU1, SBU2, SHIELD, GND, A1, A4, A5, A6, A7, A8, B5, B6, B7, B8)

USB_C_PI

C52 100n · C53 100n

U6 — USBLC6-2SC6 (ESP_DP, ESP_DM)

C54 100n

## SENSOR-STROOK (G V S PER GPIO) + I2C

J14 — SENSOR S (G0–G19)

J15 — SENSOR V (V0–V19)

J16 — SENSOR G (GND ×20)

J17 — (GND, V3V3, IMU_SCL)

J18 — (GND, V3V3, IMU_SDA, IMU_SCL)

## BEVESTIGING

H1 H2 H3 H4 — M3

## SERVO-BUS A (L1, L2, L3)

U5 — SN74AHCT1G126 (GND, VCC, TX_A, BUSA_DRV)

U8 — SN74LVC1G34 (GND, VCC, BUSA_RXIN, RX_A)

R51 120R (BUSA_DRV) · R52 10k (TX_A) · R53 10k (BUSA_RXIN) · R54 120R (BUSA_RXIN) · C55 100n · C56 100n

J2 — L1 servo (GND, V12, BUS_A)

J3 — L2 servo (GND, V12, BUS_A)

J4 — L3 servo (GND, V12, BUS_A)

## SERVO-BUS B (L4, L5, L6)

U7 — SN74AHCT1G126 (GND, VCC, TX_B, BUSB_DRV)

U9 — SN74LVC1G34 (GND, VCC, BUSB_RXIN, RX_B)

R55 120R (BUSB_DRV) · R56 10k (TX_B) · R57 10k (BUSB_RXIN) · R58 120R (BUSB_RXIN) · C57 100n · C58 100n

J5 — L4 servo (GND, V12, BUS_B)

J6 — L5 servo (GND, V12, BUS_B)

J7 — L6 servo (GND, V12, BUS_B)

ARACHNE-18 MINI BRAIN v1.0 – ESP32-S3 + MPU-6050 hexapod servo controller for a Raspberry Pi 4 (generated by tools/build_sch.py from tools/design.py)
Servo connectors: 1 = GND, 2 = +12 V, 3 = DATA (Feetech STS half-duplex bus, 1 Mbaud). CHECK the pin order against your cable before plugging in!
Bus A = legs L1 L2 L3 (ESP32 UART1: TX IO17, RX IO18, DE IO16). Bus B = legs L4 L5 L6 (UART2: TX IO10, RX IO11, DE IO9). DE = RTS in RS485 half-duplex mode.
Pi 4 link: USB-C (board) <-> USB-A (Pi) cable on the ESP32-S3 native USB -> /dev/ttyACM0 on the Pi. Pi VBUS only feeds the ESP32 (through D7), never the servos.
5V Pi pads: buck output (5 V, 3 A for the Pi 4) -> Pi GPIO pin 2/4 (+5V) and 6 (GND), or a cut USB-C lead into the Pi power socket.
Buck = TI TPS54560B datasheet design (fig. 32): 400 kHz, 5.0 V 5 A, UVLO start 6.5 V.

ESP32-S3 + MPU-6050, 6 servo-bus legs, USB to Pi 4
Kobe Vanrobaeys – maakleerplek
Sheet: /
File: arachne_mini.kicad_sch
Title: ARACHNE-18 mini brain board
Size: A1 | Date: 2026-10-18 | Rev: 1.0
KiCad E.D.A. 10.0.3 | Id: 1/1